Yunkyeong Kang

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Closing the Design–Build–Test–Learn loop for enzyme discovery, with GMP quality assurance experience from a commercial biologics plant.

PROFILE

Bioengineering researcher working on two sides of the same industry. In research, I combine protein language model embeddings with cell-free, biofoundry-scale experimental validation to find new enzymes in metagenomes, and I build both the embedding pipelines and the assays that test them. Before graduate school I spent 1 year 8 months in GMP quality assurance at a biologics plant, supporting MFDS and FDA inspections as an English interpreter, investigating deviations and managing change control. I am looking for positions in AI-driven bio R&D as well as in quality and regulatory affairs, including RA documentation and R&D planning. Business-level English (OPIc IH) with regulatory-inspection interpreting experience. Available from January 2027.

RESEARCH INTERESTS

Protein language model embeddings · Metagenomic enzyme discovery · Cell-free protein synthesis and biofoundry automation · Design–Build–Test–Learn and active-learning loops · Structure-based interpretation of function · GMP quality systems and regulated evidence

EDUCATION

M.S. in Engineering Biology — KAIST, Graduate School of Engineering Biology

Feb 2025 – Feb 2027 (expected) · Advisor: Prof. Haseong Kim · KAIST–KRIBB joint program

Thesis: *Improving the Efficiency of Metagenomic Enzyme Screening with an Embedding-Based Iterative Learning Strategy*

B.S. in Biological Science — Sungkyunkwan University

Mar 2016 – Feb 2023 · Magna Cum Laude · major GPA 4.31 / 4.5

Thesis: *Plant Breeding Techniques for Crop Improvement: Focusing on Genome Editing*

RESEARCH EXPERIENCE

Graduate Researcher — Synthetic Biology and Bioengineering Research Center, KRIBB

Feb 2025 – present, Daejeon · KAIST–KRIBB joint graduate program

Building an iterative Design–Build–Test–Learn workflow for enzyme discovery from metagenomes, in which protein language model embeddings select candidates and cell-free protein synthesis on a biofoundry measures what they do.

- Narrowed ~5.3M ORFs from 16 environmental metagenomes to a tractable candidate set, recovering sequences too distant for homology search to return, by generating ProtT5 embeddings as a 1,024-dimensional vector database and ranking by cosine similarity to the mean embedding of 98 structurally characterized bacterial lipases.
- Raised cell-free expression yield ~2.5× over a commercial kit by GFP comparison, and detected pNPO/pNPB/pNPP activity in the CalB control that the kit had not, by preparing S30 extract in-house from a GroEL/GroES-overexpressing strain; shortened library construction by carrying the T7 promoter/terminator in the primer 5′ tails, so that rounds run under identical conditions.
- Identified 43 hit candidates (5.0%), 16 of them active on two or more substrates, by screening 855 candidate ORFs from 9 libraries against 3 pNP substrates on a 96/384-well automated workflow (Echo 525 acoustic dispenser, automated liquid handlers) and replacing ad-hoc thresholds with a robust Z-score criterion ($Z \geq 3$ plus positive-control comparison).
- Making the selection criterion itself update each round, by training an activity prediction model on the Round 1 measurements (frozen ESM-C encoder with trainable pooling and projection heads) in which the ORFs with no detected activity serve as experimentally demonstrated negatives rather than negatives guessed from sequence distance.
- Structural interpretation of hits with ESMFold and Foldseek fold-similarity search, plus FoldDisco analysis of the discontinuous catalytic motif (Ser–His–Asp triad, oxyanion hole).
- Digitized the lab notebook and built a reagent, instrument and sample database so that experimental records are consistent enough to be reused as data.

Undergraduate Researcher — Research Center for Plant Plasticity, Seoul National University

Dec 2021 – Feb 2022

Temperature-dependent protein changes in *Arabidopsis*; yeast two-hybrid, insert cloning, restriction digestion. Wrote a lab-specific protocol still used as training material.

Undergraduate Researcher — Infection and Inflammatory Disease Lab, Sungkyunkwan University

Sep 2021 – Dec 2021

Immune-cell changes in diet-induced obese C57BL/6 mice; cell culture, ELISA, western blot, flox genotyping.

INDUSTRY EXPERIENCE

Quality Assurance, Biologics — Hanmi Pharmaceutical, Pyeongtaek Bio Plant

Jun 2023 – Feb 2025 (1 yr 8 mo), full-time

Biologics quality assurance group at a commercial GMP plant producing pre-filled-syringe biologics, covering regulatory inspections, deviations and change control, and supplier quality.

- Supported MFDS and FDA inspections as English interpreter and scribe, including supply-chain lines of questioning, and prepared the documentation requested by inspectors.
- Investigated manufacturing deviations, identified root causes and managed the associated change control.
- Managed supplier qualification and supply-chain quality documentation under GMP.

PUBLICATIONS & PRESENTATIONS

Global Representations in Overlooked Single-token Embeddings of Protein Language Models. S. Heo, **Y. Kang**, G. Park, S.-G. Lee, D.-H. Lee, H. Kim. Poster Q-12, 52nd Annual Meeting of the Korean Society for Microbiology and Biotechnology (KMB 2025), Busan, Jun 2025.

Showed that a single-token embedding can represent whole-protein information, improving bacterial essential-gene prediction by up to 17% over average pooling.

SELECTED PROJECTS

MetaForge — Deep Learning-Based Discovery of Moonlighting Enzymes

IDEA-B Synthetic Biology Competition (MSIT / Korea Synthetic Biology Council), Mar – Oct 2026 · team of 6, candidate selection and validation design

Passed first-round review and was funded as a commissioned KRIBB project. Established the rationale from literature and database review, narrowed 14 enzyme–substrate reactions to 7 through pre-screening controls, then screened 380 metagenomic candidates × 7 substrates × 4 conditions (~10,000 conditions) in three days.

MAKEAT — AI Diet-Tracking App

K-Software Empowerment BootCamp (640 h), 2022–2023 · Excellence Award, healthcare track

Team project; volunteered for the Android/Flutter development role in addition to database management. Individual projects: oral medication image classifier, COVID-19 case data analysis.

LLM agents for research and workflow automation

Personal projects, 2025 – present · repositories private, walkthrough available on request

Built agents for the repetitive parts of research: paper translation and summary with a fixed glossary, meeting minutes from recordings, a lab notebook assistant that expands a short plan into a templated record, and a job-posting digest that collects, filters and mails a daily report unattended in CI. Each is a Claude Code project with its own skills, commands and hooks.

DIFFERENTIATING STRENGTHS

- **Dry–wet integration.** I build the embedding pipelines and run the assays that test them, so prediction and measurement are designed against each other rather than handed over a wall.
- **Industry quality experience.** 1 yr 8 mo of GMP quality assurance, including MFDS and FDA inspection support, applied to how research evidence is documented and controlled.
- **Automation mindset.** From biofoundry liquid handling to LLM agents, I move repetitive work into systems that run the same way every time.

SKILLS

Computational

Python (PyTorch), R, Linux/GPU server workflows; protein language models (ProtT5, ESM-C, ESM-2) and embedding database construction (HDF5); ESMFold, AlphaFold3, RFdiffusion, LigandMPNN; Foldseek, FoldDisco; UMAP and contrastive representation learning; NGS/multi-omics analysis.

Experimental

Cell-free protein synthesis and S30 extract preparation; primer design, PCR, Gibson assembly, transformation; enzyme activity assays (pNP substrates), SDS-PAGE; cell culture, ELISA, western blot, yeast two-hybrid.

Automation & instruments

Echo 525 acoustic dispenser, Zephyr G3 liquid handler, Trobot II automated PCR, 5400 Fragment Analyzer, microplate readers (Victor Nivo, Tecan Spark).

Quality & regulatory

GMP; regulatory inspection support (MFDS, FDA); deviation investigation and root cause analysis; change control; supplier and supply-chain quality management; technical documentation in Korean and English.

Languages

Korean (native); English (OPIc IH, 2026; TOEIC 960, 2021 — certificate expired; elementary schooling in San Jose, California; regulatory inspection interpreting experience).

AWARDS & ACTIVITIES

- Magna Cum Laude, Sungkyunkwan University (Feb 2023)
- Academic Excellence Scholarship, highest tier, Sungkyunkwan University (2022)
- Excellence Award, Healthcare Track, K-Software Empowerment BootCamp (Feb 2023)
- KAIST Language Buddy Program, participant (Sep 2025 – Aug 2026) — all team coordination in English
- WISET Junior Mentor, STEM career mentoring for high school students (2017)